Nesterov: Smooth minimization of non-smooth functions

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Abstract

This report studies Nesterov's smoothing technique for structured non-smooth convex optimization problems. The method applies to objectives whose non-smooth part can be written as a maximization over an auxiliary variable. This maximization is regularized by a strongly convex prox-function, producing a smooth approximation with an explicit error bound. The resulting smooth problem is solved by an accelerated first-order method, and convergence is measured by a primal-dual gap certificate.

The implementation is tested on matrix games, the continuous location problem, and two absolute-value objectives: the maximum of absolute values and the sum of absolute values. The experiments reproduce the qualitative behavior of Nesterov's matrix game example, compare several problem classes, examine the empirical dependence of the iteration count on the accuracy parameter ε, and test a continuation strategy in which the smoothing parameter is decreased during optimization.

Contents

1 Introduction

Many convex optimization problems are non-smooth because their objectives contain maxima, norms, or absolute values. Such functions are still convex, but standard gradient methods cannot be applied directly, and classical subgradient methods usually have a slow worst-case convergence rate. Nesterov's smoothing method addresses this difficulty for a special but important class of non-smooth problems: those whose non-smooth term has an explicit maximization structure.

The central idea is to replace the auxiliary maximization by a regularized maximization. The regularization is controlled by a smoothing parameter μ. A larger value of μ gives a smoother and easier problem, while a smaller value gives a more accurate approximation of the original non-smooth objective. This creates a trade-off between approximation accuracy and the conditioning of the smoothed problem.

The aim of this report is to implement and experimentally study this smoothing framework. First, the theoretical construction from Nesterov's paper is summarized: the structured problem class, the smooth approximation, the accelerated method, the primal-dual gap certificate, and the resulting $O(1/\varepsilon)$ complexity bound. Next, the problem classes used in the experiments are introduced. Finally, computational experiments are presented for matrix games, the continuous location problem, the maximum of absolute values problem, and the sum of absolute values problem.

The experiments focus on certified convergence. Instead of reporting only objective values, the stopping criterion is the primal-dual gap, which gives an explicit certificate that the returned solution is ε-optimal. The experiments also compare the theoretical iteration prediction with observed iteration counts and test whether a practical continuation strategy for the smoothing parameter can improve performance.

2 Theoretical Background

This section describes the theoretical framework used in the implementation. The method applies to structured non-smooth convex problems whose non-smooth part is written as a maximization over an auxiliary variable. Nesterov's smoothing technique regularizes this maximization with a strongly convex prox-function. The result is a differentiable approximation with a controlled approximation error. The smoothed problem is then minimized by an accelerated first-order method, and the quality of the computed solution is certified by a primal-dual gap.

2.1 Problem class

Nesterov's smoothing technique [1] applies to convex problems of the form

$$f^* = \min_{x \in Q_1} f(x),$$

where Q_1 is a closed convex set. The important assumption is that the objective has the structured form

$$f(x) = \hat{f}(x) + \max_{u \in Q_2} \left\{ \langle Ax, u \rangle - \hat{\phi}(u) \right\}.$$

Here Q_2 is another closed convex set, A is a linear operator, $\hat{f}$ is a smooth convex function, and $\hat{\phi}$ is a convex function on Q_2. The non-smoothness comes from the maximization over the auxiliary variable u. In the examples considered in this report, $\hat{f} = 0$, so the objective is generated entirely by the max-term.

The variable x is the primal variable. The variable u is the auxiliary, or dual, variable. For a fixed x, it identifies the active part of the maximum and is later used to construct the dual certificate.

2.2 Prox-functions and smoothing

The method uses prox-functions on the primal and auxiliary feasible sets. Let d_1 be a prox-function on Q_1, strongly convex with parameter $\sigma_1 > 0$, and let d_2 be a prox-function on Q_2, strongly convex with parameter $\sigma_2 > 0$. As in the paper, the prox-functions are assumed to be normalized so that their minimum values on the corresponding sets are zero. Define

$$D_1 = \max_{x \in Q_1} d_1(x), \qquad D_2 = \max_{u \in Q_2} d_2(u).$$

For a smoothing parameter $\mu > 0$, the max-term is replaced by

$$f_\mu(x) = \max_{u \in Q_2} \left\{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \right\}.$$

The smooth approximation of the full objective is therefore

$$\bar{f}_\mu(x) = \hat{f}(x) + f_\mu(x).$$

The additional term $-\mu d_2(u)$ makes the inner maximization strongly concave in the dual variable and gives a differentiable approximation.

The approximation error is uniformly bounded:

$$\bar{f}_\mu(x) \le f(x) \le \bar{f}_\mu(x) + \mu D_2.$$

Thus μD_2 is an upper bound on the smoothing error. Smaller μ gives a more accurate approximation, but it also makes the smoothed function less smooth.

2.3 Gradient of the smoothed objective

For each x, define the smoothed dual response

$$u_\mu(x) = \arg\max_{u \in Q_2} \left\{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \right\}.$$

Nesterov shows that the smoothed max-term is differentiable and

$$\nabla f_\mu(x) = A^* u_\mu(x).$$

Consequently,

$$\nabla \bar{f}_\mu(x) = \nabla \hat{f}(x) + A^* u_\mu(x).$$

If M is the Lipschitz constant of $\nabla \hat{f}$, then the gradient of $\bar{f}_\mu$ is Lipschitz continuous with constant

$$L_\mu = M + \frac{\|A\|_{1,2}^2}{\mu \sigma_2}.$$

This formula is central to the method: decreasing μ improves the approximation error, but increases L_μ, so the smooth problem becomes harder for a first-order method.

2.4 Accelerated minimization of the smoothed problem

For a the standard method, the algorithm minimizes

$$\min_{x \in Q_1} \bar{f}_\mu(x).$$

The accelerated scheme used in the paper maintains three sequences x_k, y_k, z_k. At iteration k, it computes

$$g_k = \nabla \bar{f}_\mu(x_k).$$

The local point is

$$y_k = T_{Q_1}(x_k),$$

where

$$T_{Q_1}(x_k) = \arg\min_{y \in Q_1} \left\{ \langle g_k, y - x_k \rangle + \frac{L}{2}\|y - x_k\|^2 \right\}.$$

The method also forms an aggregate point from all previous gradients. With

$$\alpha_k = \frac{k+1}{2}, \qquad G_k = \sum_{i=0}^{k} \alpha_i g_i,$$

define

$$z_k = \arg\min_{x \in Q_1} \left\{ \frac{L}{\sigma_1} d_1(x) + \langle G_k, x \rangle \right\}.$$

The next point is the convex combination

$$x_{k+1} = \frac{2}{k+3} z_k + \frac{k+1}{k+3} y_k.$$

This is the accelerated part of the method. In the implementation, the formulas above are specialized to the geometry of each problem class.

2.5 Dual averaging and stopping criterion

At iteration k, the method computes the smoothed dual response

$$u_k = u_\mu(x_k).$$

The final dual candidate is formed as a weighted average of these responses:

$$\hat{u} = \frac{\sum_{k=0}^{N} \alpha_k u_k}{\sum_{k=0}^{N} \alpha_k}, \qquad \alpha_k = \frac{k+1}{2}.$$

Equivalently,

$$\hat{u} = \sum_{k=0}^{N} \frac{2(k+1)}{(N+1)(N+2)} u_k.$$

Since Q_2 is convex and each $u_k \in Q_2$, the averaged point also satisfies $\hat{u} \in Q_2$.

After N iterations, the final primal point is taken to be the last y-iterate,

$$\hat{x} = y_N.$$

The corresponding dual value is defined by

$$\phi(u) = \min_{x \in Q_1} \left\{ \hat{f}(x) + \langle Ax, u \rangle \right\} - \hat{\phi}(u).$$

By weak duality,

$$\phi(\hat{u}) \le f^* \le f(\hat{x}).$$

Therefore,

$$f(\hat{x}) - \phi(\hat{u})$$

is a primal-dual gap. In the implementation, this quantity is used as the stopping criterion. The algorithm stops when

$$f(\hat{x}) - \phi(\hat{u}) \le \varepsilon.$$

This gives a certificate that the computed point is ε-optimal.

2.6 Complexity bound

Nesterov's analysis chooses the smoothing parameter as a function of the planned number of iterations. For a fixed number N, the smoothing parameter is chosen as

$$\mu(N) = \frac{2\|A\|_{1,2}}{N+1}\sqrt{\frac{D_1}{\sigma_1\sigma_2 D_2}}.$$

With this choice, the method gives the primal-dual estimate

$$0 \le f(\hat{x}) - \phi(\hat{u}) \le \frac{4\|A\|_{1,2}}{N+1}\sqrt{\frac{D_1 D_2}{\sigma_1\sigma_2}} + \frac{4MD_1}{\sigma_1(N+1)^2}.$$

Here M is the Lipschitz constant of $\nabla\hat{f}$. In the examples implemented in this project, $\hat{f} = 0$, so $M = 0$. The bound then becomes

$$f(\hat{x}) - \phi(\hat{u}) \le \frac{4\|A\|_{1,2}}{N+1}\sqrt{\frac{D_1 D_2}{\sigma_1\sigma_2}}.$$

If the required accuracy is fixed in advance, this estimate can be rewritten in terms of ε. To guarantee

$$f(\hat{x}) - \phi(\hat{u}) \le \varepsilon,$$

it is sufficient to take

$$N + 1 \ge \frac{4\|A\|_{1,2}}{\varepsilon}\sqrt{\frac{D_1 D_2}{\sigma_1\sigma_2}}.$$

Substituting this value of $N+1$ into $\mu(N)$ gives the equivalent accuracy-based choice

$$\mu = \frac{\varepsilon}{2D_2}.$$

Thus the smoothing error satisfies

$$\mu D_2 = \frac{\varepsilon}{2}.$$

For this choice of μ, the Lipschitz constant of the smoothed objective is

$$L_\mu = M + \frac{\|A\|_{1,2}^2}{\mu\sigma_2} = M + \frac{2D_2}{\sigma_2}\frac{\|A\|_{1,2}^2}{\varepsilon}.$$

When $M = 0$, this simplifies to

$$L_\mu = \frac{2D_2}{\sigma_2}\frac{\|A\|_{1,2}^2}{\varepsilon}.$$

The resulting theoretical iteration complexity is therefore

$$N = O\left(\frac{1}{\varepsilon}\right).$$

This is the predicted iteration bound used later in the computational experiments.

3 Problem Classes

This section presents the problem classes used in the experiments. All of them are based on application examples discussed by Nesterov. The goal is to show how each problem can be written in the structured form

$$\min_{x\in Q_1}\left\{\hat{f}(x) + \max_{u\in Q_2}\left(\langle Ax, u\rangle - \hat{\phi}(u)\right)\right\}.$$

In all examples considered here, $\hat{f}(x) = 0$. Thus, the essential step is to express the non-smooth objective as a maximization over an auxiliary variable.

3.1 Matrix games

The matrix game problem is

$$\min_{x\in\Delta_n}\max_{u\in\Delta_m} u^\top Ax,$$

where $A\in\mathbb{R}^{m\times n}$, and Δ_n, Δ_m are probability simplexes. This problem can be interpreted as a two-player zero-sum game. The vector x is the mixed strategy of the minimizing player, while u is the mixed strategy of the maximizing player.

The problem already has the required max-structure, because

$$u^\top Ax = \langle Ax, u\rangle.$$

Thus,

$$Q_1 = \Delta_n, \qquad Q_2 = \Delta_m, \qquad \hat{\phi}(u) = 0.$$

The final form is

$$\boxed{\min_{x\in\Delta_n}\max_{u\in\Delta_m} \langle Ax, u\rangle.}$$

Equivalently,

$$\max_{u\in\Delta_m} u^\top Ax = \max_{1\le j\le m} (Ax)_j,$$

so the objective is the maximum of finitely many linear functions.

3.2 Continuous location problem

The continuous location problem is

$$\min_{\|x\|_2\le r} \sum_{j=1}^{p} w_j\|x-c_j\|_2.$$

Here x is the location of a facility, c_j are fixed city or customer locations, and $w_j>0$ are weights. The objective is the total weighted distance from the facility to the given points.

The Euclidean norm has the representation

$$\|x-c_j\|_2 = \max_{\|u_j\|_2\le 1} \langle x-c_j, u_j\rangle.$$

Applying this to each distance term gives

$$\sum_{j=1}^{p} w_j\|x-c_j\|_2 = \max_{\|u_j\|_2\le 1}\left\{\sum_{j=1}^{p} w_j\langle x, u_j\rangle - \sum_{j=1}^{p} w_j\langle c_j, u_j\rangle\right\}.$$

The primal feasible set is

$$Q_1 = \{x : \|x\|_2 \le r\}.$$

The auxiliary variable is $U=(u_1,\dots,u_p)$, with

$$Q_2 = \{U=(u_1,\dots,u_p) : \|u_j\|_2\le 1,\ j=1,\dots,p\}.$$

With the operator

$$\mathcal{A}x = (w_1x,\dots,w_px),$$

we have

$$\langle \mathcal{A}x, U\rangle = \sum_{j=1}^{p} w_j\langle x, u_j\rangle.$$

Therefore,

$$\hat{\phi}(U) = \sum_{j=1}^{p} w_j \langle c_j, u_j \rangle.$$

The final form is

$$\boxed{\min_{\|x\|_2 \le r} \max_{\|u_j\|_2 \le 1} \left\{ \sum_{j=1}^{p} w_j \langle x, u_j \rangle - \sum_{j=1}^{p} w_j \langle c_j, u_j \rangle \right\}.}$$

3.3 Maximum of absolute values problem

The maximum of absolute values problem is

$$\min_{\|x\|_2 \le r} \max_{1 \le j \le m} |a_j^\top x - b_j|.$$

This problem minimizes the worst absolute residual among the affine expressions $a_j^\top x - b_j$.

Let

$$A = \begin{bmatrix} a_1^\top \\ \vdots \\ a_m^\top \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}.$$

Using

$$|a_j^\top x - b_j| = \max\{a_j^\top x - b_j,\ -a_j^\top x + b_j\},$$

define

$$\widetilde{A} = \begin{bmatrix} A \\ -A \end{bmatrix}, \qquad \widetilde{b} = \begin{bmatrix} -b \\ b \end{bmatrix}.$$

Then

$$\max_{1 \le j \le m} |a_j^\top x - b_j| = \max_{1 \le \ell \le 2m} (\widetilde{A}x + \widetilde{b})_\ell.$$

A maximum of coordinates can be written as a maximization over the simplex:

$$\max_{1 \le \ell \le 2m} (\widetilde{A}x + \widetilde{b})_\ell = \max_{u \in \Delta_{2m}} u^\top (\widetilde{A}x + \widetilde{b}).$$

Thus,

$$Q_1 = \{x : \|x\|_2 \le r\}, \qquad Q_2 = \Delta_{2m}, \qquad \hat{f}(x) = 0, \qquad \hat{\phi}(u) = -\widetilde{b}^\top u.$$

The final form is

$$\boxed{\min_{\|x\|_2 \le r} \max_{u \in \Delta_{2m}} \left\{ \langle \widetilde{A}x, u \rangle + \widetilde{b}^\top u \right\}.}$$

3.4 Sum of absolute values problem

The sum of absolute values problem is

$$\min_{\|x\|_2 \le r} \sum_{j=1}^{m} |a_j^\top x - b_j|.$$

This problem minimizes the total absolute residual over all affine expressions $a_j^\top x - b_j$.

The identity

$$|t| = \max_{|u| \le 1} ut$$

gives

$$|a_j^\top x - b_j| = \max_{|u_j|\le 1} u_j(a_j^\top x - b_j).$$

Since the variables u_j are chosen independently,

$$\sum_{j=1}^m |a_j^\top x - b_j| = \max_{\|u\|_\infty \le 1} \sum_{j=1}^m u_j(a_j^\top x - b_j).$$

Expanding the expression gives

$$\sum_{j=1}^m u_j(a_j^\top x - b_j) = \langle Ax, u\rangle - b^\top u.$$

Thus,

$$Q_1 = \{x : \|x\|_2 \le r\}, \qquad Q_2 = \{u : \|u\|_\infty \le 1\} = [-1,1]^m, \qquad \hat{f}(x) = 0, \qquad \hat{\phi}(u) = b^\top u.$$

The final form is

$$\boxed{\min_{\|x\|_2\le r} \max_{\|u\|_\infty\le 1} \left\{\langle Ax, u\rangle - b^\top u\right\}.}$$

4 Changing the Smoothing Parameter

The smoothing parameter controls the trade-off between accuracy and smoothness. For a prescribed accuracy ε, the implementation of Nesterov's smoothing method uses

$$\mu_\varepsilon = \frac{\varepsilon}{2D_2}.$$

This choice makes the smoothing error bounded by

$$\mu_\varepsilon D_2 = \frac{\varepsilon}{2}.$$

The remaining part of the error is then controlled by the accelerated minimization of the smoothed objective.

When high accuracy is required, μ_ε becomes small. This gives a sharper approximation of the original objective, but it also increases the Lipschitz constant

$$L_\mu = M + \frac{\|A\|_{1,2}^2}{\mu\sigma_2}.$$

Thus the smoothed problem can become harder to minimize.

This motivates a continuation strategy. The final accuracy requirement is not changed, but the method reaches the final smoothing parameter gradually. It starts from a larger value

$$\mu_0 = c\mu_\varepsilon, \qquad c > 1,$$

and decreases the smoothing parameter geometrically,

$$\mu_{t+1} = \rho\mu_t, \qquad 0 < \rho < 1,$$

until μ_ε is reached. At each stage, the accelerated method is applied to the smoothed problem with the current value μ_t. The transition to the next stage is made after the current primal-dual gap reaches a prescribed intermediate tolerance, for example

$$f(\hat{x}) - \phi(\hat{u}) \le c_{\text{stage}}\mu_t D_2,$$

where $c_{\text{stage}} > 0$ determines how accurately the intermediate stage is solved. At the last stage, the usual final condition is imposed:

$$f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon.$$

The continuation strategy is a practical modification of the standard smoothing method. It is tested in this report to see whether progress made on smoother intermediate problems can reduce the total computational effort. The concrete values of c, ρ, and c_{stage} are specified in the experimental section.

5 Experiments and Results

This section presents the computational experiments. The experiments are divided into four parts. First, the matrix game experiment from Nesterov's paper is reproduced and used as the reference test. Second, the same experimental scheme is applied to the other problem classes introduced above. Third, a fixed-size experiment is used to study the dependence of the iteration count on the accuracy parameter ε. Finally, the standard smoothing method is compared with the continuation strategy described in Section 4.

In all experiments, convergence is measured by the primal-dual gap

$$f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon.$$

Thus the reported stopping condition is a certificate of ε-optimality, not only a numerical decrease in the objective value.

5.1 Reproduction of Nesterov's matrix game experiment

The first experiment reproduces the matrix game experiment from Nesterov's paper. The problem is

$$\min_{x \in \Delta_n} \max_{u \in \Delta_m} u^\top A x,$$

where $A \in \mathbb{R}^{m \times n}$. Random matrices were generated with entries sampled uniformly from $[-1, 1]$.

The tested grid was

ε	m	n
10^{-2}	$100, 300, 1000$	$100, 300, 1000, 3000, 10000$
10^{-3}	$100, 300, 1000$	$100, 300, 1000, 3000, 10000$
10^{-4}	$100, 300, 1000$	$100, 300, 1000, 3000$

Similarly to the original paper, the results are presented as separate tables for each value of ε. Rows correspond to m, columns correspond to n, and each cell contains

iterations / percentage of predicted iterations.

$m \backslash n$	100	300	1000	3000	10000
100	765 / 42%	934 / 46%	1 108 / 49%	1 187 / 49%	1 281 / 49%
300	690 / 34%	910 / 40%	1 126 / 45%	1 268 / 47%	1 475 / 51%
1000	761 / 34%	873 / 35%	1 086 / 39%	1 284 / 43%	1 457 / 46%

Table 1: Matrix game reproduction results for $\varepsilon = 0.01$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	100	300	1000	3000	10000
100	7 123 / 39%	8 426 / 41%	8 928 / 40%	10 191 / 42%	10 080 / 39%
300	7 337 / 36%	8 639 / 38%	10 154 / 40%	11 302 / 42%	12 022 / 41%
1000	7 526 / 33%	8 828 / 35%	10 157 / 37%	11 595 / 39%	14 167 / 44%

Table 2: Matrix game reproduction results for $\varepsilon = 0.001$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	100	300	1000	3000
100	68 505 / 37%	64 512 / 31%	77 136 / 34%	74 024 / 30%
300	60 932 / 30%	78 680 / 34%	83 844 / 33%	92 862 / 34%
1000	70 778 / 31%	77 644 / 31%	93 346 / 34%	102 925 / 35%

Table 3: Matrix game reproduction results for $\varepsilon = 0.0001$. Each cell shows iterations / percentage of predicted iterations.

The aggregate results are summarized in Table 4.

ε	Instances	Avg. iter.	Max iter.	Avg. % pred.	Max gap	Avg. time
10^{-4}	12	78 766	102 925	33.00%	0.000100	465s
10^{-3}	15	9 765	14 167	39.07%	0.001000	140s
10^{-2}	15	1 080	1 475	43.13%	0.009997	6.83s

Table 4: Aggregate summary of the matrix game reproduction experiment.

Summary of results. The matrix game experiment reproduces the qualitative behavior reported by Nesterov. All tested instances reached the requested primal-dual gap. The number of iterations was consistently below the theoretical worst-case prediction, but the scaling with respect to ε was clearly visible: reducing ε by one order of magnitude increased the iteration count by approximately one order of magnitude. The results therefore support the $O(1/\varepsilon)$ dependence while also showing that the theoretical bound is conservative on these random instances.

5.2 Experiments for the other examples

After reproducing the matrix game experiment, analogous experiments were performed for the other implemented examples. The goal was to check whether the same smoothing framework behaves similarly for different non-smooth structures.

The tested problems were:

1. continuous location problem,
2. maximum of absolute values problem,
3. sum of absolute values problem.

In each case, the same type of information was recorded: number of iterations, predicted number of iterations, percentage of the predicted bound used, runtime, and final primal-dual gap.

5.2.1 Continuous location problem

The continuous location problem is

$$\min_{\|x\|_2 \le r} \sum_{j=1}^{p} w_j \|x - c_j\|_2.$$

City locations were generated randomly inside a Euclidean ball and positive weights were assigned to the cities. The radius was fixed to $r = 1$.

The tested grid was

ε	number of cities	dimension	r
10^{-2}	$10, 50$	$2, 5$	1
10^{-3}	$10, 50$	$2, 5$	1
10^{-4}	$10, 50$	$2, 5$	1

The results are presented separately for each value of ε. Rows correspond to the number of cities p, columns correspond to the dimension d, and each cell contains

iterations / percentage of predicted iterations.

$p \backslash d$	2	5
10	566 / 35%	932 / 47%
50	3 606 / 35%	2 847 / 29%

Table 5: Continuous location results for $\varepsilon = 0.01$. Each cell shows iterations / percentage of predicted iterations.

$p \backslash d$	2	5
10	10 890 / 62%	11 407 / 49%
50	31 653 / 33%	28 151 / 30%

Table 6: Continuous location results for $\varepsilon = 0.001$. Each cell shows iterations / percentage of predicted iterations.

$p \backslash d$	2	5
10	48 121 / 28%	75 747 / 36%
50	320 867 / 34%	241 660 / 25%

Table 7: Continuous location results for $\varepsilon = 0.0001$. Each cell shows iterations / percentage of predicted iterations.

The aggregate results are shown in Table 8.

ε	Instances	Avg. iter.	Max iter.	Avg. % pred.	Max gap	Avg. time
10^{-4}	4	171 599	320 867	30.64%	0.000100	31.8s
10^{-3}	4	20 525	31 653	43.74%	0.001000	3.86s
10^{-2}	4	1 988	3 606	36.70%	0.009997	0.35s

Table 8: Summary of continuous location experiments.

5.2.2 Maximum of absolute values problem

The maximum of absolute values problem is

$$\min_{\|x\|_2 \le r} \max_{1 \le j \le m} |a_j^\top x - b_j|.$$

Random matrices A and vectors b were generated with entries sampled uniformly from $[-1, 1]$. The radius was fixed to $r = 1$.

The tested grid was

ε	m	n	r
10^{-2}	$100, 300$	$50, 200$	1
10^{-3}	$100, 300$	$50, 200$	1
10^{-4}	$100, 300$	$50, 200$	1

The results are presented separately for each value of ε. Rows correspond to m, columns correspond to n, and each cell contains

iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	1 793 / 60%	3 018 / 53%
300	1 773 / 51%	3 788 / 60%

Table 9: Maximum of absolute values results for $\varepsilon = 0.01$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	16 673 / 54%	13 649 / 24%
300	16 135 / 48%	33 671 / 53%

Table 10: Maximum of absolute values results for $\varepsilon = 0.001$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	152 449 / 49%	116 635 / 20%
300	167 001 / 50%	249 861 / 40%

Table 11: Maximum of absolute values results for $\varepsilon = 0.0001$. Each cell shows iterations / percentage of predicted iterations.

The aggregate results are shown in Table 12.

ε	Instances	Avg. iter.	Max iter.	Avg. % pred.	Max gap	Avg. time
10^{-4}	4	171 486	249 861	39.81%	0.000100	10.6s
10^{-3}	4	20 032	33 671	44.76%	0.001000	1.26s
10^{-2}	4	2 593	3 788	56.06%	0.009998	0.16s

Table 12: Summary of maximum of absolute values experiments.

5.2.3 Sum of absolute values problem

The sum of absolute values problem is

$$\min_{\|x\|_2 \le r} \sum_{j=1}^{m} |a_j^\top x - b_j|.$$

Random matrices A and vectors b were generated with entries sampled uniformly from $[-1, 1]$. The radius was fixed to $r = 1$.

The tested grid was

ε	m	n	r
10^{-2}	$100, 300$	$50, 200$	1
10^{-3}	$100, 300$	$50, 200$	1
10^{-4}	$100, 300$	$50, 200$	1

The results are presented separately for each value of ε. Rows correspond to m, columns correspond to n, and each cell contains

iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	5 146 / 27%	5 561 / 21%
300	27 750 / 58%	9 896 / 16%

Table 13: Sum of absolute values results for $\varepsilon = 0.01$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	40 986 / 22%	61 784 / 23%
300	284 262 / 60%	156 544 / 25%

Table 14: Sum of absolute values results for $\varepsilon = 0.001$. Each cell shows iterations / percentage of predicted iterations.

$m\backslash n$	50	200
100	677 811 / 34%	140 591 / 5%
300	2 711 368 / 58%	1 263 846 / 20%

Table 15: Sum of absolute values results for $\varepsilon = 0.0001$. Each cell shows iterations / percentage of predicted iterations.

The aggregate results are shown in Table 16.

ε	Instances	Avg. iter.	Max iter.	Avg. % pred.	Max gap	Avg. time
10^{-4}	4	1 198 404	2 711 368	29.45%	0.000100	54.9s
10^{-3}	4	135 894	284 262	32.52%	0.001000	6.43s
10^{-2}	4	12 088	27 750	30.43%	0.010000	0.51s

Table 16: Summary of sum of absolute values experiments.

Summary of results. The additional experiments show that the same smoothing framework works beyond matrix games. All tested instances of the continuous location problem, the maximum of absolute values problem, and the sum of absolute values problem reached the required primal-dual gap. As in the matrix game experiment, the actual iteration counts were below the theoretical predictions.

The results also show that the difficulty of the problem depends strongly on the structure of the non-smooth term. The continuous location and maximum of absolute values problems were relatively cheaper for the tested sizes. The sum of absolute values problem required many more iterations, which is consistent with the fact that this objective aggregates all residuals rather than selecting only the largest one.

5.3 Fixed-size epsilon-scaling experiment

The previous experiments used grids with several problem dimensions. To examine the dependence on ε more directly, an additional fixed-size experiment was performed.

For each problem class and each random seed, one random instance was generated and reused for all values of ε. Thus, only the accuracy parameter changed. This isolates the dependence of the iteration count on ε.

The tested values were

$$\varepsilon \in \{10^{-1}, 5 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}, 10^{-3}, 5 \cdot 10^{-4}, 10^{-4}\}.$$

The fixed problem sizes were:

Problem	Size
Matrix game	$m = 300,\ n = 1000$
Continuous location	$p = 50,\ d = 5,\ r = 1$
Maximum of absolute values	$m = 300,\ n = 200,\ r = 1$
Sum of absolute values	$m = 300,\ n = 200,\ r = 1$

The numerical results are summarized in Table 17.

Problem	ε	$1/\varepsilon$	Mean iter.	Std iter.	Mean gap	Max gap
Matrix game	10^{-1}	10	79	8.44	0.099210	0.099693
Matrix game	$5 \cdot 10^{-2}$	20	213	5.70	0.049820	0.049973
Matrix game	10^{-2}	100	1 097	12.52	0.009987	0.009996
Matrix game	$5 \cdot 10^{-3}$	200	2 141	35.85	0.004997	0.005000
Matrix game	10^{-3}	1 000	9 806	212.18	0.001000	0.001000
Matrix game	$5 \cdot 10^{-4}$	2 000	18 634	513.54	0.000500	0.000500
Matrix game	10^{-4}	10 000	82 064	2 622	0.000100	0.000100
Continuous location	10^{-1}	10	337	18.26	0.099485	0.099937
Continuous location	$5 \cdot 10^{-2}$	20	662	37.22	0.049880	0.049966
Continuous location	10^{-2}	100	3 338	300.01	0.009996	0.009999
Continuous location	$5 \cdot 10^{-3}$	200	6 633	609.15	0.004998	0.004999
Continuous location	10^{-3}	1 000	33 049	3 085	0.001000	0.001000
Continuous location	$5 \cdot 10^{-4}$	2 000	66 469	5 902	0.000500	0.000500
Continuous location	10^{-4}	10 000	332 524	29 668	0.000100	0.000100
Maximum of absolute values	10^{-1}	10	445	5.93	0.099316	0.099969
Maximum of absolute values	$5 \cdot 10^{-2}$	20	841	5.26	0.049825	0.049918
Maximum of absolute values	10^{-2}	100	3 830	22.48	0.009998	0.010000
Maximum of absolute values	$5 \cdot 10^{-3}$	200	7 453	24.63	0.004999	0.005000
Maximum of absolute values	10^{-3}	1 000	34 107	357.33	0.001000	0.001000
Maximum of absolute values	$5 \cdot 10^{-4}$	2 000	63 575	1 016	0.000500	0.000500
Maximum of absolute values	10^{-4}	10 000	240 017	9 058	0.000100	0.000100
Sum of absolute values	10^{-1}	10	1 562	89.13	0.099690	0.099962
Sum of absolute values	$5 \cdot 10^{-2}$	20	2 813	286.48	0.049985	0.049999
Sum of absolute values	10^{-2}	100	12 791	1 782	0.010000	0.010000
Sum of absolute values	$5 \cdot 10^{-3}$	200	25 048	3 830	0.005000	0.005000
Sum of absolute values	10^{-3}	1 000	122 206	19 730	0.001000	0.001000
Sum of absolute values	$5 \cdot 10^{-4}$	2 000	242 824	40 122	0.000500	0.000500
Sum of absolute values	10^{-4}	10 000	1 207 464	201 485	0.000100	0.000100

Table 17: Summary of the fixed-size epsilon-scaling experiment.

5.3.1 Iterations versus $1/\varepsilon$

The first plot shows the mean number of iterations as a function of $1/\varepsilon$.

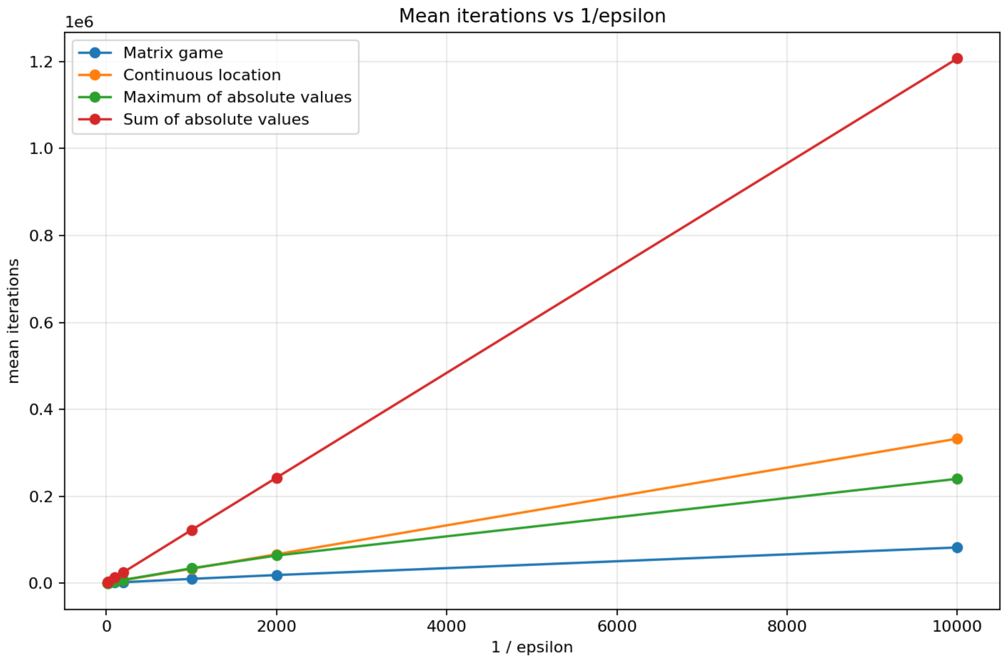

Figure 1: Mean number of iterations versus $1/\varepsilon$.

This plot gives a direct visual check of the theoretical prediction. If

$$N(\varepsilon) \approx C\frac{1}{\varepsilon},$$

then the curves should be approximately linear in $1/\varepsilon$. This is what is observed for all four problem classes. The main difference between the curves is the constant factor: the sum of absolute values problem requires the most iterations, while the matrix game is the cheapest in this fixed-size test. The overall shape is consistent with the expected complexity

$$N = O(1/\varepsilon).$$

5.3.2 Log-log plot and fitted exponent

To examine the empirical rate more precisely, the iteration counts were also analyzed on a log-log scale. If the number of iterations satisfies a relation of the form

$$N(\varepsilon) \approx C\varepsilon^{-p},$$

then taking logarithms gives

$$\log N = \log C + p \log(1/\varepsilon).$$

Thus, the exponent p can be estimated as the slope of a linear regression between $\log N$ and $\log(1/\varepsilon)$. The theoretical complexity $N = O(1/\varepsilon)$ corresponds to

$$p = 1.$$

The log-log plot is shown in Figure 2. The approximately linear shape of the curves confirms that a power-law model is a reasonable description of the observed dependence on ε.

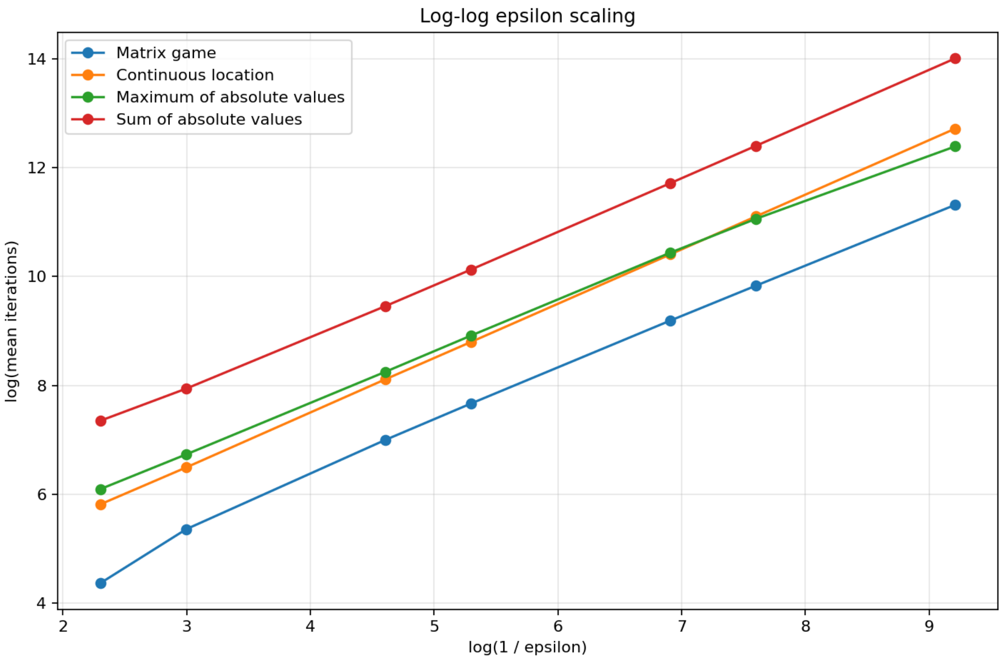

Figure 2: Log-log plot for the epsilon-scaling experiment.

The fitted exponents are shown in Table 18. The table reports the estimated exponent $\hat{p}$ and its standard error.

Problem	$\hat{p}$	SE
Matrix game	0.9897	0.0096
Continuous location	0.9986	0.0054
Maximum of absolute values	0.9208	0.0042
Sum of absolute values	0.9649	0.0096

Table 18: Fitted exponents in the empirical model $N(\varepsilon) \approx C\varepsilon^{-p}$.

The fitted exponents for the matrix game and continuous location problem are very close to 1, which is consistent with the theoretical $O(1/\varepsilon)$ complexity. For the maximum of absolute values problem and the sum of absolute values problem, the fitted exponents are slightly below 1. This means that, for these random instances, the observed iteration counts grew slightly slower than the worst-case $1/\varepsilon$ rate.

Overall, the log-log analysis supports the theoretical prediction. The observed rates are close to $1/\varepsilon$, and in some cases slightly better. This is consistent with the interpretation that the theoretical complexity bound is a worst-case estimate and may be conservative on randomly generated instances.

5.4 Experiment with changing the smoothing parameter μ

The previous experiments used Nesterov's smoothing method directly with

$$\mu_\varepsilon = \frac{\varepsilon}{2D_2}.$$

This experiment compares that standard choice with the continuation strategy from Section 4.

For the continuation runs, the smoothing parameter starts at

$$\mu_0 = 8\mu_\varepsilon$$

and is then updated by

$$\mu_{t+1} = 0.5\mu_t$$

until μ_ε is reached. The tested sequence is therefore

$$8\mu_\varepsilon, \quad 4\mu_\varepsilon, \quad 2\mu_\varepsilon, \quad \mu_\varepsilon.$$

An intermediate stage is stopped when

$$f(\hat{x}) - \phi(\hat{u}) \leq \mu_t D_2,$$

and the final stage is stopped when

$$f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon.$$

The experiment was performed for all four problem classes, using the same fixed problem sizes as in the epsilon-scaling experiment:

Problem	Size
Matrix game	$m = 300,\ n = 1000$
Continuous location	$p = 50,\ d = 5,\ r = 1$
Maximum of absolute values	$m = 300,\ n = 200,\ r = 1$
Sum of absolute values	$m = 300,\ n = 200,\ r = 1$

The tested accuracies were

$$\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}.$$

For each problem, each value of ε, and each random seed, the same random instance was solved twice: once with the standard smoothing method and once with the continuation strategy. This makes the comparison fair, since both runs solve the same instance.

5.4.1 Iteration comparison

Table 19 reports the average number of iterations for the standard smoothing method and for continuation.

Problem	ε	μ mode	Mean iter.	Std iter.
Continuous location	10^{-4}	Continuation	379 813	37 174
Continuous location	10^{-4}	Fixed μ	306 630	29 498
Continuous location	10^{-3}	Continuation	37 330	10 378
Continuous location	10^{-3}	Fixed μ	30 237	8 283
Continuous location	10^{-2}	Continuation	2 923	1 247
Continuous location	10^{-2}	Fixed μ	3 240	483.64
Matrix game	10^{-4}	Continuation	115 059	2 313
Matrix game	10^{-4}	Fixed μ	88 760	1 864
Matrix game	10^{-3}	Continuation	12 853	416.61
Matrix game	10^{-3}	Fixed μ	9 997	326.88
Matrix game	10^{-2}	Continuation	1 466	29.78
Matrix game	10^{-2}	Fixed μ	1 126	30.57
Maximum of absolute values	10^{-4}	Continuation	331 092	26 995
Maximum of absolute values	10^{-4}	Fixed μ	260 380	18 927
Maximum of absolute values	10^{-3}	Continuation	44 448	536.44
Maximum of absolute values	10^{-3}	Fixed μ	34 493	385.07
Maximum of absolute values	10^{-2}	Continuation	5 059	68.72
Maximum of absolute values	10^{-2}	Fixed μ	3 790	81.63
Sum of absolute values	10^{-4}	Continuation	1 679 708	193 437
Sum of absolute values	10^{-4}	Fixed μ	1 297 689	147 250
Sum of absolute values	10^{-3}	Continuation	150 866	7 829
Sum of absolute values	10^{-3}	Fixed μ	115 690	6 887
Sum of absolute values	10^{-2}	Continuation	18 853	2 015
Sum of absolute values	10^{-2}	Fixed μ	14 448	1 536

Table 19: Comparison of the standard smoothing method and continuation.

The results show that continuation did not reduce the iteration count in most cases. For the matrix game, the maximum of absolute values problem, and the sum of absolute values problem, continuation required more iterations than the standard method for all tested accuracies. The same pattern appears for the continuous location problem at $\varepsilon = 10^{-3}$ and $\varepsilon = 10^{-4}$. The only average improvement occurred for the continuous location problem at $\varepsilon = 10^{-2}$.

For the tested instances and continuation parameters, the benefit of starting with a smoother problem did not compensate for the extra work required by the additional stages.

5.4.2 Iteration speedup

To make the comparison clearer, Table 20 reports the iteration speedup

$$\text{speedup} = \frac{\text{iterations for the standard method}}{\text{iterations for continuation}}.$$

Thus, values greater than 1 mean that continuation used fewer iterations, while values below 1 mean that the standard method was better.

Problem	ε	Mean iter. speedup	Min iter. speedup	Max iter. speedup
Continuous location	10^{-4}	0.807	0.806	0.810
Continuous location	10^{-3}	0.811	0.806	0.818
Continuous location	10^{-2}	1.28	0.804	1.89
Matrix game	10^{-4}	0.771	0.768	0.774
Matrix game	10^{-3}	0.778	0.772	0.782
Matrix game	10^{-2}	0.768	0.759	0.778
Maximum of absolute values	10^{-4}	0.787	0.775	0.792
Maximum of absolute values	10^{-3}	0.776	0.770	0.781
Maximum of absolute values	10^{-2}	0.749	0.740	0.759
Sum of absolute values	10^{-4}	0.773	0.764	0.779
Sum of absolute values	10^{-3}	0.767	0.756	0.778
Sum of absolute values	10^{-2}	0.766	0.758	0.773

Table 20: Iteration speedup of continuation relative to the standard smoothing method.

The speedup table confirms the previous observation. Most speedup values are below 1, usually around 0.75–0.81. This means that continuation typically required about 20%–30% more iterations than the standard method. The only average speedup above 1 occurs for the continuous location problem with $\varepsilon = 10^{-2}$, where the mean speedup is approximately 1.28. However, even in this case the minimum speedup is below 1, so the improvement was not consistent across all seeds.

Overall, this experiment suggests that the standard smoothing method is more efficient for the tested settings.

5.4.3 Gap history

In addition to final iteration counts, the primal-dual gap was tracked during optimization for representative runs with

$$\varepsilon = 10^{-4}, \qquad \text{seed} = 0.$$

This gives more detailed information about the behavior of both methods. The following plots show the primal-dual gap as a function of the iteration number. The y-axis is logarithmic, and the horizontal dotted line marks the target accuracy $\varepsilon = 10^{-4}$. For the continuation method, vertical dotted lines indicate changes of the smoothing parameter μ.

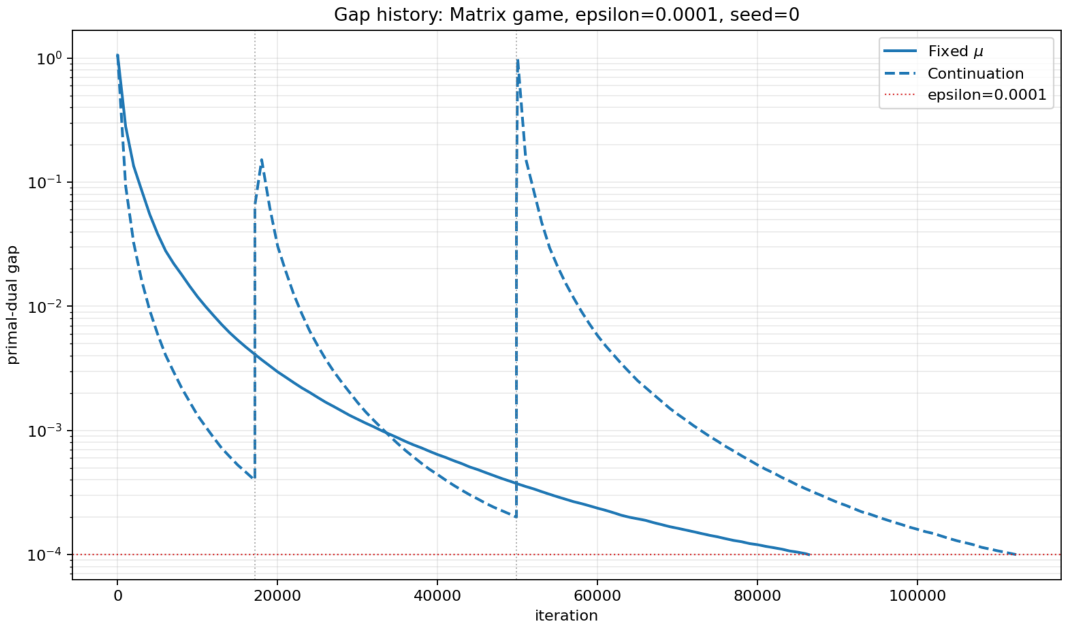

Figure 3: Gap history for the matrix game with $\varepsilon = 10^{-4}$.

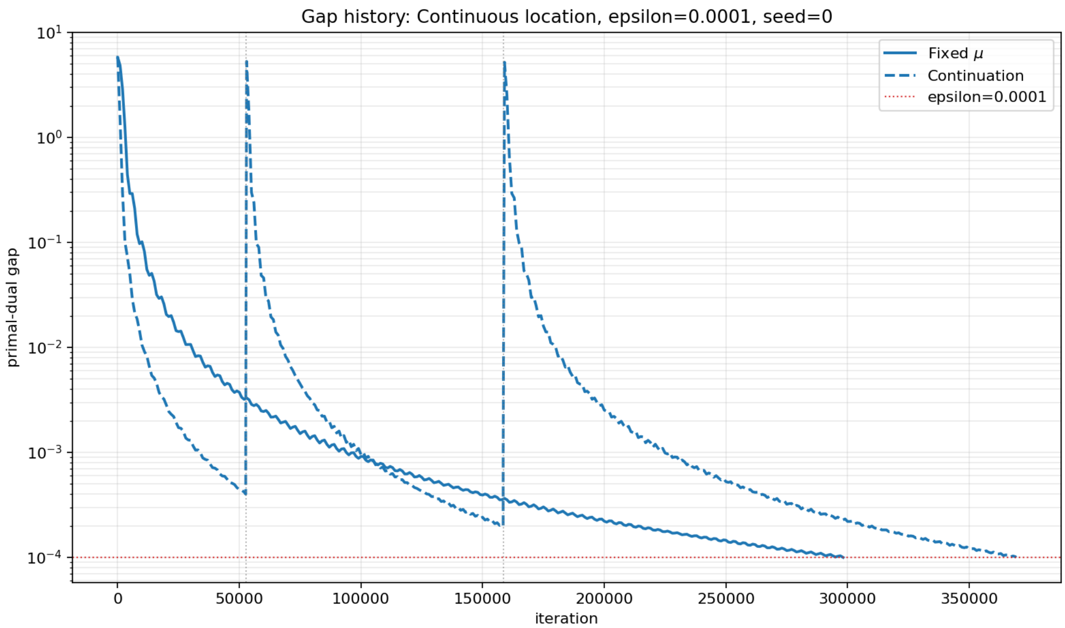

Figure 4: Gap history for the continuous location problem with $\varepsilon = 10^{-4}$.

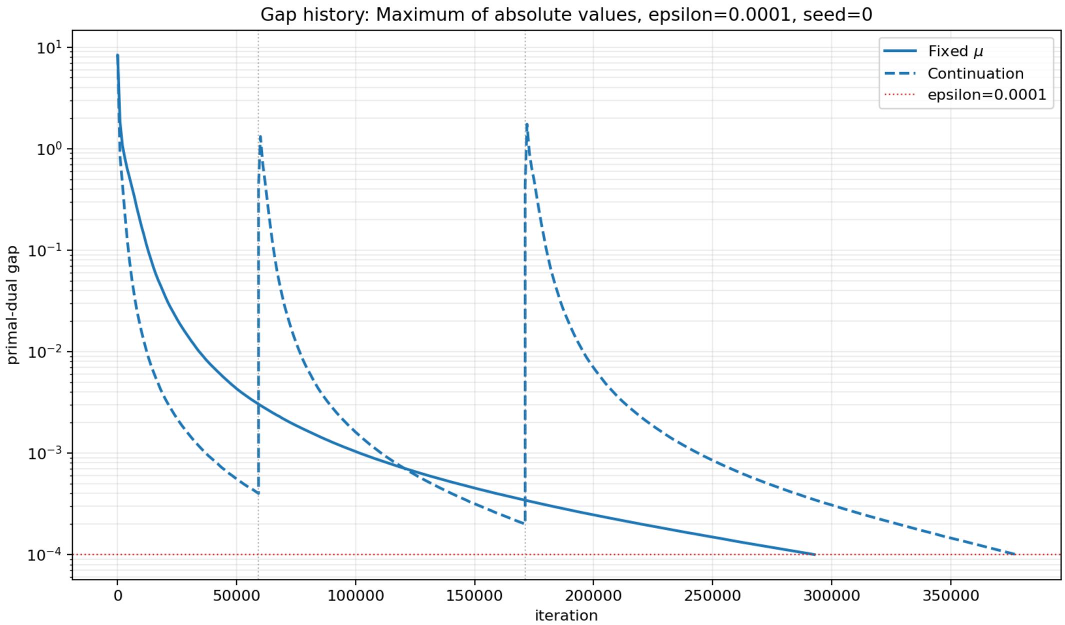

Figure 5: Gap history for the maximum of absolute values problem with $\varepsilon = 10^{-4}$.

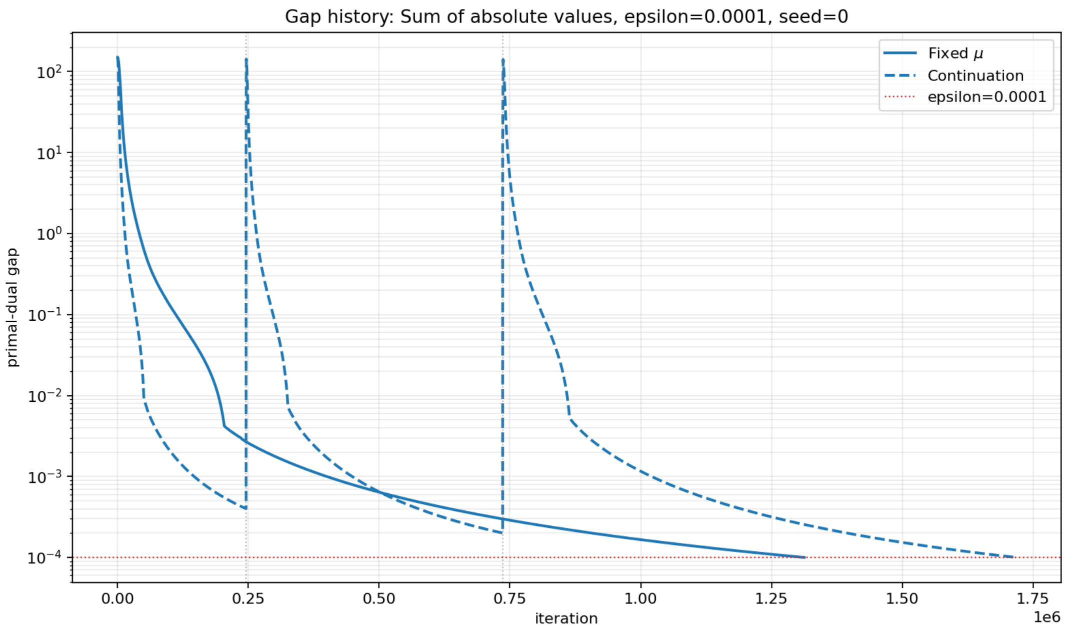

Figure 6: Gap history for the sum of absolute values problem with $\varepsilon = 10^{-4}$.

The plots show that continuation reduces the gap quickly inside each stage, but after each decrease of μ, the gap increases again. This is expected: changing μ changes the smoothed objective and makes the approximation closer to the original non-smooth function. Therefore, the current point may no longer have as small a gap for the new, less smoothed problem.

For the tested instances, the standard smoothing method reaches the target line earlier than continuation. This is especially clear for the matrix game, maximum of absolute values problem, and sum of absolute values problem. In the continuous location problem, the difference is smaller, but the standard method still reaches the final tolerance earlier for the displayed $\varepsilon = 10^{-4}$ run.

6 Discussion and conclusion

The experiments support the main theoretical message of Nesterov's smoothing method. For all tested reproduction-style instances, the algorithm reached the required primal-dual gap. This means that the stopping criterion provided a certificate of solution quality. At the same time, the observed iteration counts were substantially smaller than the worst-case predictions. This is natural, since the theoretical bound is designed for the worst case, while the numerical tests use randomly generated instances.

The fixed-size epsilon-scaling experiment gives the clearest evidence for the predicted dependence on the accuracy parameter. For the matrix game and the continuous location problem, the fitted exponents were very close to 1, in agreement with the theoretical $O(1/\varepsilon)$ complexity. For the maximum of absolute values and the sum of absolute values problems, the fitted exponents were slightly below 1. This does not contradict the theorem; it only means that the observed growth was somewhat better than the worst-case rate on these instances.

The comparison between problem classes shows that the hidden constants are important in practice. The sum of absolute values problem required many more iterations than the other examples, even though its empirical dependence on ε remained close to linear. This reflects the structure of the objective. The maximum of absolute values problem depends on the largest residual, whereas the sum of absolute values problem aggregates all residuals. This can lead to a larger effective Lipschitz constant after smoothing and therefore to a larger number of iterations.

The continuation experiment shows that changing the smoothing parameter is not automatically beneficial. Although larger values of μ make the early smoothed problems easier, the method must solve several stages. In addition, when μ is reduced, the smoothed approximation changes and the primal-dual gap can increase. With the tested schedule $\mu_0 = 8\mu_\varepsilon$ and decay factor 0.5, the standard smoothing method was usually more efficient. Continuation should therefore be treated as a heuristic whose performance depends on the chosen schedule and intermediate stopping rules.

Overall, the implementation confirms that Nesterov's smoothing framework is a useful approach for structured non-smooth convex optimization. It gives a systematic way to replace a non-smooth max-term by a smooth approximation, apply an accelerated first-order method, and stop with a certified primal-dual gap. The experiments are consistent with the theoretical $O(1/\varepsilon)$ rate and show that the method is effective for several different problem structures. They also show that practical performance depends on the objective structure and on the choice of smoothing strategy.

References

[1] Yurii Nesterov. Smooth minimization of non-smooth functions. *Mathematical Programming*, 103(1):127–152, 2005.